

Estimated Annual Agricultural Pesticide Use Benton County, Washington 2014-2018

24 July 2026

Summary

Benton County ranks **#2 of Washington’s 39 counties** by total estimated agricultural pesticide mass applied (EPest-high method, summed 2014-2018), making it one of WA’s five highest pesticide-use counties. Benton County, part of the Tri-Cities area, is a center of Washington’s wine industry (Yakima Valley and Columbia Valley AVAs), alongside tree fruit and irrigated row crops.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Benton County, Washington** during 2018, as reported by [USGS Estimated Annual Agricultural Pesticide Use](#) data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound’s chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by [Thelin and Stone \(2013\)](#) and [Baker and Stone \(2015\)](#):

“EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD.” -[Pesticide National Synthesis Project webpage](#)

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. “detailed interpretation of where and how much use occurs within a county is not appropriate”).
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)
(continued)

Code	Chemical class
OTH	Other
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Benton County, WA, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	METAM	DTC	2599681	1788427	1659330
2	CHLOROPICRIN	OC	670204	247415	164239
3	BACILLUS AMYLOLIQUEFACIEN	MIC	539737	258537	166573
4	METAM POTASSIUM	DTC	513582	407422	442950
5	PETROLEUM OIL	OTH	443422	465297	406179
6	DICHLOROPROPENE	OC	386646	625036	862660
7	SULFUR	INO	277318	203567	189476
8	SULFURIC ACID	INO	183378	386776	390333
9	GLYPHOSATE	OTH	97737	115152	92511
10	EPTC	TC	40426	27414	25697
11	BURKHOLDERIA SPP	MIC	38036	33439	25114
12	2,4-D	CPX	30516	33322	28274
13	CHLOROTHALONIL	OTH	29301	19233	16627
14	PARAQUAT	DIP	28981	20954	16149
15	GLUFOSINATE	OTH	28776	14365	10688
16	MANCOZEB	DTC	27696	35539	37198
17	CALCIUM POLYSULFIDE	INO	27463	57700	53408
18	DCPA	OTH	27348	39362	35665
19	COPPER	INO	18808	15690	12561
20	METHOMYL	CB	16391	19663	19452
21	OXAMYL	CB	15220	5913	12069
22	COPPER HYDROXIDE	INO	12283	12028	16386
23	CHLORPYRIFOS	OP	12052	13097	15506
24	METOLACHLOR & METOLACHLOR-S	OTH	12031	11644	11832
25	METOLACHLOR-S	OTH	12031	11644	11787
26	DIMETHENAMID & DIMETHENAMID-P	OTH	11985	8652	7747
27	DIMETHENAMID-P	OTH	11985	8652	7747
28	KAOLIN CLAY	INO	9739	9212	8070
29	COPPER OXYCHLORIDE	INO	8918	7735	5521
30	PENDIMETHALIN	OTH	7190	14110	17407
31	DIQUAT	DIP	7177	2900	2490
32	ATRAZINE	TRZ	7062	4952	3675
33	METRIBUZIN	TRZ	6901	3872	4200
34	DIMETHOATE	OP	6295	4653	4208
35	FLUTOLANIL	OTH	6162	3398	3391
36	LINURON	OTH	5845	3113	1996
37	BROMOXYNIL	OTH	5844	7912	7819
38	FLUAZINAM	OTH	5168	3052	2919
39	MALEIC HYDRAZIDE	OTH	4882	5890	5414
40	MEFENOXAM	OTH	4556	3043	2553
41	BOSCALID	OTH	4199	4702	5250
42	CARBARYL	CB	4106	4271	5547
43	MCPA	CPX	4100	6096	5295
44	PHOSPHORIC ACID	INO	3803	4084	15250
45	POTASSIUM BICARBONATE	INO	3170	5502	4145
46	ETHOFUMESATE	OTH	3126	2246	1728

Table 2: Top 80 EPest-high Estimates, Benton County, WA, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) *(continued)*

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
47	BENTAZONE	OTH	2846	2955	2886
48	PYRACLOSTROBIN	OTH	2687	3312	3158
49	MCPB	CPX	2583	1398	848
50	TRIFLUMIZOLE	OTH	2512	2705	2958
51	CYPRODINIL	OTH	2257	2330	1908
52	PROPARGITE	OTH	2190	2082	2588
53	TRIFLURALIN	OTH	2087	1827	1521
54	SPIROMESIFEN	OTH	2055	1588	1412
55	FLUMIOXAZIN	OTH	1874	1018	666
56	PYMETROZINE	OTH	1867	1021	988
57	CYMOXANIL	OTH	1836	2016	1971
58	AZOXYSTROBIN	OTH	1705	3410	3180
59	SPIROTETRAMAT	OTH	1587	1935	1776
60	THIOPHANATE-METHYL	OTH	1555	2479	2996
61	BIFENTHRIN	PYR	1467	1126	943
62	PENTHIOPYRAD	OTH	1441	2185	1977
63	FLUOPYRAM	OTH	1438	1274	1106
64	TOLFENPYRAD	OTH	1420	1420	1420
65	DIFENOCONAZOLE	OTH	1356	1140	928
66	BUPROFEZIN	OTH	1352	1619	1122
67	PROPICONAZOLE	OTH	1298	2532	2092
68	FAMOXADONE	OTH	1256	1749	1773
69	TRIFLOXYSTROBIN	OTH	1168	951	958
70	DICAMBA	OTH	1148	3048	2484
71	BACILLUS THURINGIENSIS	MIC	1108	1915	1378
72	CYANTRANILIPROLE	OTH	1051	923	810
73	CHLORANTRANILIPROLE	OTH	957	773	807
74	DIMETHOMORPH	OTH	918	683	551
75	SPINOSYN	OTH	894	1286	1437
76	SPINETORAM	OTH	862	1007	1214
77	COPPER SULFATE	INO	834	4822	4373
78	CLOPYRALID	OTH	815	988	799
79	NORFLURAZON	OTH	809	294	296
80	IMIDACLOPRID	OTH	803	1390	2202

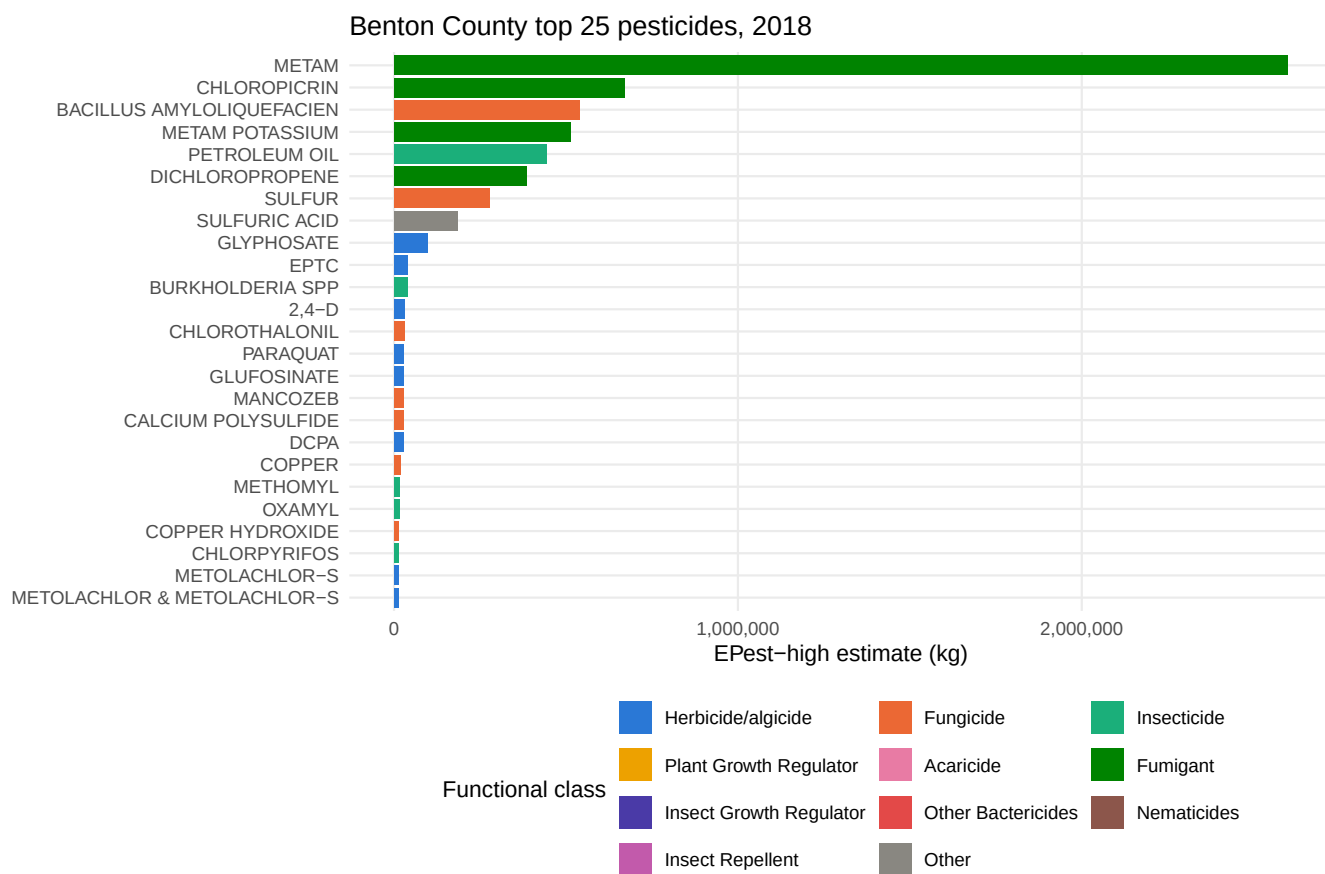


Figure 1: Top 25 compounds in Benton County by E Pest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Benton County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	METAM	493256	2599681
2	CHLOROPICRIN	0	670204
3	BACILLUS AMYLOLIQUEFACIEN	0	539737
4	METAM POTASSIUM	3335	513582
5	PETROLEUM OIL	443422	443422
6	DICHLOROPROPENE	218850	386646
7	SULFUR	276922	277318
8	SULFURIC ACID	0	183378
9	GLYPHOSATE	89049	97737
10	EPTC	2779	40426
11	BURKHOLDERIA SPP	921	38036
12	2,4-D	30473	30516
13	CHLOROTHALONIL	401	29301
14	PARAQUAT	18928	28981
15	GLUFOSINATE	28590	28776
16	MANCOZEB	27692	27696
17	CALCIUM POLYSULFIDE	27463	27463
18	DCPA	27348	27348
19	COPPER	18808	18808
20	METHOMYL	10839	16391
21	OXAMYL	0	15220
22	COPPER HYDROXIDE	4288	12283
23	CHLORPYRIFOS	12052	12052
24	METOLACHLOR & METOLACHLOR-S	588	12031
25	METOLACHLOR-S	588	12031
26	DIMETHENAMID & DIMETHENAMID-P	7375	11985
27	DIMETHENAMID-P	7375	11985
28	KAOLIN CLAY	9739	9739
29	COPPER OXYCHLORIDE	2079	8918
30	PENDIMETHALIN	6002	7190
31	DIQUAT	0	7177
32	ATRAZINE	3218	7062
33	METRIBUZIN	1157	6901
34	DIMETHOATE	326	6295
35	FLUTOLANIL	0	6162
36	LINURON	386	5845
37	BROMOXYNIL	3662	5844
38	FLUAZINAM	0	5168
39	MALEIC HYDRAZIDE	0	4882
40	MEFENOXAM	1652	4556
41	BOSCALID	3779	4199
42	CARBARYL	4106	4106
43	MCPA	0	4100
44	PHOSPHORIC ACID	0	3803
45	POTASSIUM BICARBONATE	3170	3170
46	ETHOFUMESATE	0	3126
47	BENTAZONE	2846	2846
48	PYRACLOSTROBIN	2074	2687
49	MCPB	2583	2583
50	TRIFLUMIZOLE	2512	2512

Table 4: Top 50 Benton County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	1080050	1788427
2	DICHLOROPROPENE	512933	625036
3	PETROLEUM OIL	380800	465297
4	METAM POTASSIUM	1112	407422
5	SULFURIC ACID	0	386776
6	BACILLUS AMYLOLIQUEFACIEN	9038	258537
7	CHLOROPICRIN	4284	247415
8	SULFUR	197759	203567
9	GLYPHOSATE	108830	115152
10	CALCIUM POLYSULFIDE	57700	57700
11	DCPA	39362	39362
12	MANCOZEB	35203	35539
13	BURKHOLDERIA SPP	448	33439
14	2,4-D	32831	33322
15	PROPAMOCARB HCL	0	27738
16	EPTC	4365	27414
17	PARAQUAT	15890	20954
18	METHOMYL	17809	19663
19	CHLOROTHALONIL	8435	19233
20	COPPER	15690	15690
21	ETHOPROPHOS	0	14473
22	GLUFOSINATE	10865	14365
23	PENDIMETHALIN	12688	14110
24	CHLORPYRIFOS	10552	13097
25	COPPER HYDROXIDE	8114	12028
26	METOLACHLOR & METOLACHLOR-S	1865	11644
27	METOLACHLOR-S	1865	11644
28	KAOLIN CLAY	9042	9212
29	CUPROUS OXIDE	183	8862
30	DIMETHENAMID & DIMETHENAMID-P	5414	8652
31	DIMETHENAMID-P	5414	8652
32	BROMOXYNIL	5352	7912
33	COPPER OXYCHLORIDE	4342	7735
34	PERMETHRIN	0	6358
35	MCPA	3570	6096
36	OXAMYL	0	5913
37	MALEIC HYDRAZIDE	0	5890
38	DIURON	26	5605
39	POTASSIUM BICARBONATE	5502	5502
40	ATRAZINE	1564	4952
41	COPPER SULFATE	3728	4822
42	BOSCALID	3697	4702
43	DIMETHOATE	809	4653
44	CARBARYL	3550	4271
45	PHOSPHORIC ACID	1	4084
46	METRIBUZIN	1575	3872
47	CHROMOBACTERIUM SUBTUSGAE	0	3803
48	AZOXYSTROBIN	2484	3410
49	FLUTOLANIL	0	3398
50	ZINC	3314	3314

Table 5: Top 50 Benton County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	991155	1659330
2	DICHLOROPROPENE	753211	862660
3	METAM POTASSIUM	133769	442950
4	PETROLEUM OIL	355481	406179
5	SULFURIC ACID	0	390333
6	SULFUR	183860	189476
7	BACILLUS AMYLOLIQUEFACIEN	5423	166573
8	CHLOROPICRIN	2982	164239
9	GLYPHOSATE	87013	92511
10	CALCIUM POLYSULFIDE	53397	53408
11	MANCOZEB	36982	37198
12	DCPA	35665	35665
13	2,4-D	25511	28274
14	PROPAMOCARB HCL	0	27738
15	EPTC	11802	25697
16	CAPTAN	188	25338
17	BURKHOLDERIA SPP	370	25114
18	METHOMYL	17749	19452
19	PENDIMETHALIN	16335	17407
20	CHLOROTHALONIL	10144	16627
21	COPPER HYDROXIDE	13475	16386
22	PARAQUAT	12115	16149
23	CHLORPYRIFOS	12897	15506
24	PHOSPHORIC ACID	30	15250
25	COPPER	12448	12561
26	OXAMYL	8521	12069
27	METOLACHLOR & METOLACHLOR-S	2295	11832
28	METOLACHLOR-S	2293	11787
29	GLUFOSINATE	6528	10688
30	KAOLIN CLAY	7733	8070
31	BROMOXYNIL	5573	7819
32	ETHOPROPHOS	22	7750
33	DIMETHENAMID & DIMETHENAMID-P	4726	7747
34	DIMETHENAMID-P	4726	7747
35	CUPROUS OXIDE	409	7030
36	QUINTOZENE	2	5954
37	CARBARYL	5088	5547
38	COPPER OXYCHLORIDE	3271	5521
39	MALEIC HYDRAZIDE	0	5414
40	MCPA	3110	5295
41	BOSCALID	4194	5250
42	DIAZINON	894	4585
43	METIRAM	0	4570
44	COPPER SULFATE	3195	4373
45	DIMETHOATE	526	4208
46	METRIBUZIN	2563	4200
47	POTASSIUM BICARBONATE	4136	4145
48	PINOLENE	0	4141
49	DIURON	282	3979
50	ATRAZINE	1385	3675